

# Modelling Measles Using a Time-Dependent Transmission Rate

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## Abstract

Measles remains a major public health concern in Cameroon, where reported cases show a high seasonal variation that most existing models do not capture because they usually use a constant transmission rate. This study develops and evaluates a stochastic SIR model with a time-dependent transmission rate  $\beta(t)$ , using monthly WHO surveillance data from January 2012 to April 2025 to compare four candidate models combining two distributions for new infections (Poisson and Negative Binomial) with two forms of the transmission rate (constant and seasonal), with all parameters estimated using Maximum Likelihood Estimation and the Newton-Raphson algorithm because the score functions are not linear. The estimated transmission rate is higher during the dry season and lower during the rainy season, confirming that the data fluctuate according to the season of the year, and the Negative Binomial distribution gave better results than the Poisson by accounting for overdispersion. The best-performing model, the Negative Binomial with a seasonal transmission rate, achieves a Pearson chi-square statistic of  $\chi^2 = 12,134.49$ , the lowest among all four models, compared to  $\chi^2 = 20,950.50$  for the baseline Poisson model with a constant rate. These findings confirm that accounting for both seasonality and Overdispersion is essential for accurately modelling measles transmission in Cameroon and provide information for timing vaccination campaigns before the dry season.

**Keywords:** measles transmission, stochastic SIR model, time-dependent transmission rate, Negative Binomial distribution, Newton-Raphson algorithm

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